



AMIT VIKRAM REX

Curriculum Vitae

Experience

- Oct 2025– Present **Post-Doctoral Associate, Mechanical Engineering, Indian Institute of Technology, Guwahati, Assam.**
Hydrogen Assisted Cracking
Advisor: Dr. Nipal Dekka
- Feb 2025– **Project Fellow, Mechanical Engineering, Indian Institute of Technology, Patna, Bihar.**
Oct 2025 *Hygrothermal Fracture Analysis of Carbon Fiber Reinforced Polymer*
Advisor: Dr. Sunil Kumar Singh

Education

- 2019–2025 **PhD, Mechanical Engineering, Indian Institute of Technology, Patna, Bihar.**
Finite Element Analysis, Notch fatigue simulation, Fatigue and fracture mechanics (Low-cycle fatigue, High-cycle fatigue, Fatigue crack propagation), Cyclic plastic deformation modeling
- 2016–2018 **Master of Technology, Applied Mechanics, Motilal Nehru National Institute of Technology Allahabad, Prayagraj, Uttar Pradesh.**
- 2012–2016 **Bachelor of Technology, Mechanical Engineering, Dehradun Institute of Technology, Uttarakhand Technical University, Dehradun, Uttarakhand.**
- 2009–2011 **Intermediate Examination, K.R.K Maudal School, Lakhisarai, Bihar.**
- 2008–2009 **Secondary School Examination, Imperial Public School, Gopalganj, Bihar.**

Research Interest

Physics-informed modeling, Machine/deep learning approaches, Path-dependent material behavior, Fatigue and fracture mechanics, Hydrogen-assisted cracking.

Publications

Journal Articles

- 2025 **Amit Vikram Rex, Surajit Kumar Paul, Akhilendra Singh**, Effects of equi-biaxial and uni-axial tensile pre-straining paths on fatigue crack propagation behaviour of 2024-T4 aluminium alloy, *Materials Today Communications*, 49, 114062.
- 2024 **Amit Vikram Rex, Surajit Kumar Paul, Akhilendra Singh**, Assessing notch fatigue performance of aluminium alloy AA2024-T4 after uni-axial and equi-biaxial tensile pre-straining: Experimental observations and predictive models, *Theoretical and Applied Fracture Mechanics*, 131, 104402.
- 2023 **Amit Vikram Rex, Surajit Kumar Paul, Akhilendra Singh**, Influence of uniaxial and equi-biaxial tensile pre-straining on the high cycle and notch fatigue behaviour of AA2024-T4 aluminium alloy, *Theoretical and Applied Fracture Mechanics*, 128, 104176.
- 2023 **Amit Vikram Rex, Surajit Kumar Paul, Akhilendra Singh**, The influence of equi-biaxial and uniaxial tensile pre-strain on the low cycle fatigue performance of the AA2024-T4 aluminium alloy, *International Journal of Fatigue*, 173, 107699.

- 2023 **PP Dubey, Amit Vikram Rex, Abhishek Raj, Surajit Kumar Paul**, Influence of pre-strain on tensile response of extra deep drawing (EDD) steel under varying strain rates and crash performance, *Journal of Alloys and Metallurgical Systems*, 4, 100036.

[Communicated Journal Article](#)

- 2026 **Pranaw Parihar, Amit Vikram Rex, Sunil Kumar Singh**, Crack Propagation Analysis of CFRP Composites using DIC and Strain Energy Density based XIGA, *Theoretical and Applied Fracture Mechanics* (Under Review).

[Conference Proceedings](#)

- 2018 **Amit Vikram Rex, Ashutosh Kumar Upadhyay**, Mechanical Behavior of Polyurethane Foam under Quasi-Static and Repeated Impact Loading, *First Symposium and Workshop for Analytical Youth on Applied Mechanics (SWAYAM 2018)*, BITS Pilani, KK Birla Goa Campus, pp.11-12, 07/2018, Published By. Ane Books Pvt. Ltd.

Research Experience

[Indian Institute of Technology, Patna](#)

Assessing the effects of equi-biaxial and uni-axial tensile pre-straining paths on fatigue crack propagation behaviour of AA2024-T4 aluminium alloy.

Investigated the effect of tensile pre-straining on tensile properties, fracture resistance, and fatigue crack propagation using experiments and XFEM simulations. Observed minimal change in FCP rate, slight improvement in fatigue life, and a 3% increase in conditional fracture toughness (KQ).

Assessing notch fatigue performance of AA2024-T4 aluminium alloy after uni-axial and equi-biaxial tensile pre-straining.

Studied the influence of pre-straining paths on notch fatigue behaviour using a physics-based approach. Combined high-cycle fatigue experiments with the Theory of Critical Distance (TCD) and cyclic plastic zone (CPZ) concept to accurately predict fatigue life.

Analyzing the effect of tensile pre-straining on fatigue behaviour of AA2024-T4 aluminium alloy.

Evaluated the impact of uni-axial and equi-biaxial pre-straining on tensile and fatigue behaviour. EBSD analysis showed increased lattice distortion, while pre-straining enhanced tensile strength, yield stress, and fatigue strength. TCD predicted fatigue life within a ± 2 scatter band.

Assessing the effect of tensile pre-strain on the low-cycle fatigue performance of AA2024-T4 aluminium alloy.

Examined low-cycle fatigue behaviour under different pre-straining conditions by comparing fatigue life and cyclic stress-strain response. Uniaxial pre-straining improved fatigue life, while equi-biaxial pre-straining showed comparable performance to unstrained material.

Assessing the influence of pre-strain on tensile response and crash performance of extra deep drawing steel at different strain rates.

Analyzed the effect of pre-strain on tensile behaviour of EDD steel across strain rates. Observed increased strength and energy absorption with reduced elongation, and used a validated model to predict crash performance.

[Motilal Nehru National Institute of Technology Allahabad, Prayagraj](#)

Experimental study on the dynamic response of corrugated core sheet subjected to low-velocity impact.

Investigated low-velocity impact behaviour of corrugated sheets through dynamic and quasi-static experiments. Evaluated responses of filled and unfilled structures, including mechanical properties of polyurethane foam core.

Fellowships & Awards

- 2019 –2024 **Institute Fellowship** as a PhD research scholar at the Indian Institute of Technology Patna.

2016 –2018 **GATE Scholarship** of Ministry of Electronics and Information Technology (MeitY), Government of India, as an M.Tech student at the Motilal Nehru National Institute of Technology Allahabad.

Computer skills

FEA Abaqus and ANSYS APDL
Tools

CAD Design Solidworks
Software

Scientific Python, PyTorch
Computing

Position of Responsibility

2017 **Volunteer, INCAM 2017**, MNNIT Allahabad.

Teaching Assistantship

Fall 2019 **ME110: Workshop**, IIT Patna.

Spring 2020, **ME292: Measurement Laboratory**, IIT Patna.
Spring 2021

Fall 2020 **ME231: Engineering Materials Lab**, IIT Patna.

Fall 2022 **ME313: Design of Machine Elements Lab**, IIT Patna.

Fall 2022, **ME533: Finite Element Analysis Lab**, IIT Patna.
Fall, 2023

Fall 2018 **AM1153: Engineering Mechanics Lab**, MNNIT Allahabad.

Referees

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